

Situation: Write an asm programming for the following using loop:

Each lab member has 2 students. Enter from keyboard student age in decimal number (unsigned INT) and save in array age. Do following:

- Define array age as dword
- Perform addition and save to sum_age
- Display the result, and
- Query either to continue by typing “y” or exit program by typing ‘n’.

Solution:

1. Draw flow chart as in Figure 1. Steps explained in the flowchart:

1. Define age[], sum_age, mul_age: **Declare the array and variables.**
2. Input two int (unsigned) and assign to array age[]: **Input two integers and store them in the array.**
3. Perform addition and save to sum_age: **Add the two integers and store the result in sum_age.**
4. Display sum_age: **Output the result of the addition.**
5. Ask to continue by typing 'y' or exit by typing 'n': **Prompt the user to continue or exit.**
6. Continue? (y/n): **Decision point based on user input.**
7. Yes ('y'): **If 'y', repeat steps 2 to 6.**
8. No ('n'): **If 'n', end the program.**

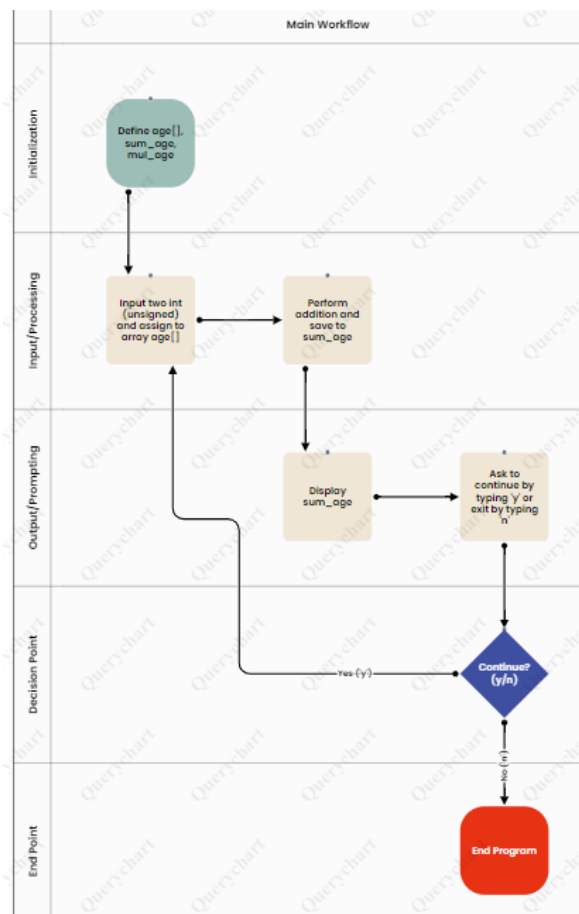


Figure 1: Add 2 number (unsigned INT)

TITLE Example Call Irvine Library and Loop

; Author: Dr Foad
; Date: 15 Jun 2024

include Irvine32.inc

.data

```
message1 BYTE "Welcome to Exercise Call Irvine Lib and Loop ", 0dh, 0ah, 0dh, 0ah
          BYTE "This program is to Add 2 input student's Age : ", 0
message2 BYTE "Input Age of Student 1 and 2 :  ", 0dh, 0ah, 0
message3 BYTE "Result Addition:  ", 0
promptBad BYTE "Invalid input, please enter again", 0
age dword 2 dup(0)
sum_age dword ?
stryn BYTE "Press 'y' to Main Menu or 'n' to Exit Program : ", 0
charIn BYTE ?
charY db 'y'
strbye BYTE "Thank you ... BYE!!", 0dh, 0ah, 0
```

.code

main **proc**

startProg:

```
    call Clrscr
    mov edx, offset message1
    call WriteString
    call crlf
    call crlf
    mov edx, offset message2
    call WriteString

    ; read input Age Students
    mov ECX, 2
    mov EBX, 0
```

loop_Age:

```
    read_inOption :
        call ReadDec
        jnc goodInOption

        mov edx, OFFSET promptBad
        call WriteString
        jmp read_inOption; go input again

    goodInOption :
        mov age[EBX], eax; store good value
        add EBX, 4

        loop loop_Age

    ; to add students age
    mov ECX, 2
    mov EBX, 0
    mov eax, 0
```

```
loopAdd:
    add eax, age[EBX]
    add ebx, 4
    loop loopAdd
```

```

; store result add
mov sum_age, eax

; display result
call crlf
mov edx, offset message3
call WriteString
mov eax, sum_age
call WriteDec
call crlf
call crlf

; ask either to continue ('y') or Exit program ('n')

mov edx, OFFSET stryn
call WriteString
call ReadChar
mov charIn, AL
call WriteChar
call Crlf
call Crlf
mov BL, charY
cmp BL, charIn
JE startProg

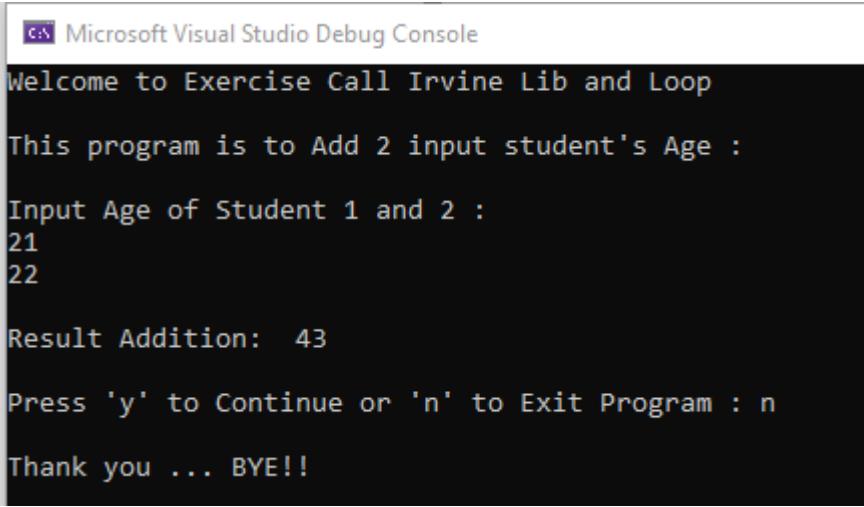
mov edx, OFFSET strbye
call WriteString

exit

main ENDP

END main

```

Sample Output:


```

Microsoft Visual Studio Debug Console

Welcome to Exercise Call Irvine Lib and Loop

This program is to Add 2 input student's Age :

Input Age of Student 1 and 2 :
21
22

Result Addition:  43

Press 'y' to Continue or 'n' to Exit Program : n

Thank you ... BYE!!

```